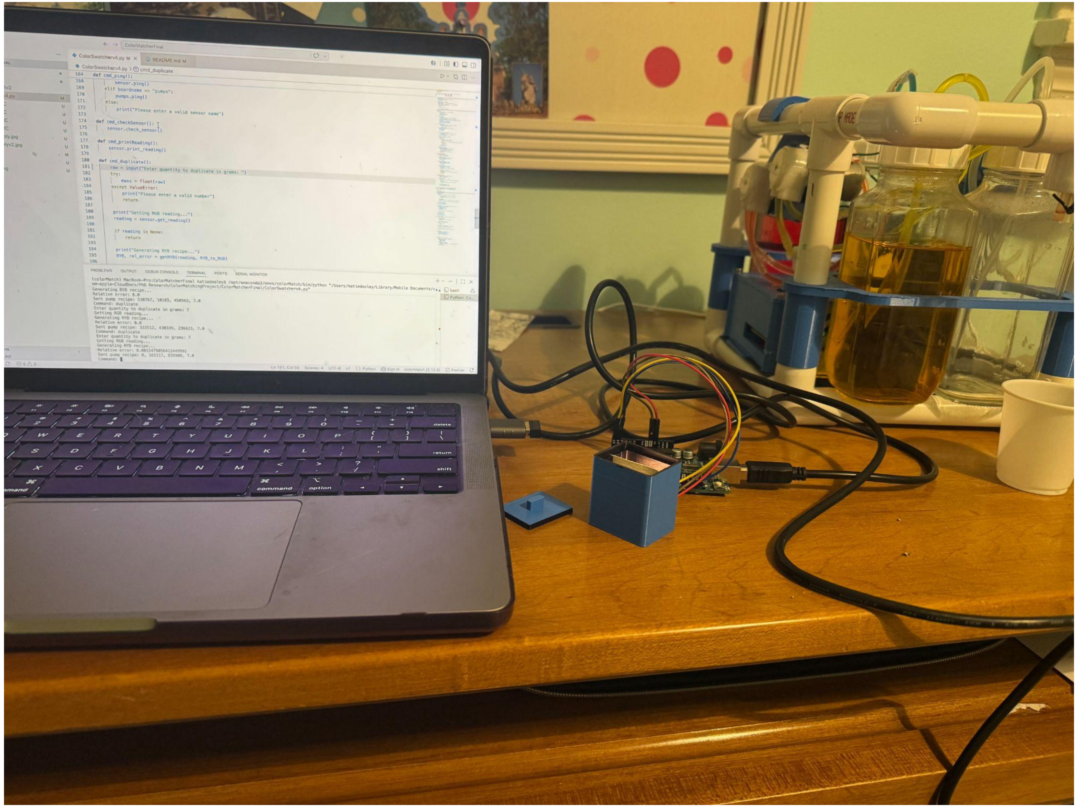
Color Swatch Duplicator Robot



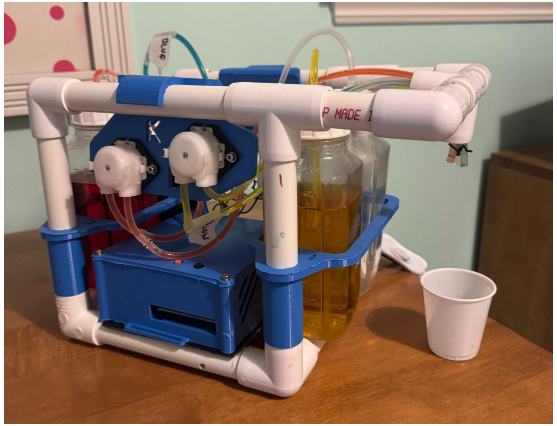
Color Swatch Duplicator

Objectives

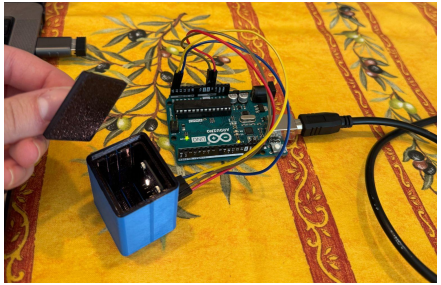
- Design a “color duplicating” robot that can analyze a color swatch and attempt to duplicate it in liquid form as close as possible with available liquid ingredients

Project Details

- Built liquid color generating system using Arduino and L298N motor drivers to control 3 peristaltic pumps
    - Designed and 3D printed structure and mountings for liquid ingredients, pumps, and electrical enclosure
- Used TCS34725 RGB color sensor installed in a 3D printed enclosure to read color samples
    - Designed enclosure to standardize readings from sensor as much as possible by fixing the position and distance of the samples from the sensor and containing the sample in a closed black box
- Programmed a Python user interface to integrate the two Arduino setups (one for the pumps and one for the sensor)



Pump Assembly



Swatch reader

Color Swatch Duplicator Results

- Ran test trying to replicate four samples (top row) using the color swatch duplicator robot
- All four samples were created by mixing colors from the original ingredients
- Robot performed well (qualitatively with the most noticeable discrepancy in the last sample (the orange)
- Duplicating algorithm is based on the “least squares” method so robot will attempt to make any color as close as possible with the available ingredients
    - Future tests: attempt to replicate colors not generated from the available ingredients and see how close the match is

- Robot control system is an open loop system because of the nature of the samples (colors are applied to paper and need to dry in order to be ready by sensor)
    - Future iterations could attempt to analyze colors when in liquid form to allow for a closed loop control system

Originals (watercolor paper dipped in dyed water)

Duplicates (generated by the robot)